



Radiation Tests Overview

- Some tests had been performed on the components by 2020, but as the new guidelines were suggesting, batch-to-batch safety factor was not applicable anymore, and each batch used on the final boards, should have been tested
- Procurement of components from single (or limited) batch(es) was started in 2021, following the initial radiation tests in 2018-2020

NIEL Tests	Unassembled components irradiated in the TRIGA reactor at JSI up to 8×10^{12} n/cm ² , followed by tests in our labs at CERN
SEE Tests	Specific designed boards for each component tested in PIF at PSI with 200 MeV proton beam
TID Tests	CMOS: Boards irradiated in CC60 facility at CERN up to 120 Gy and annealed at 100 °C (1 week) in climatic chambers at CERN
	Bipolar/Bi-CMOS: Combined with SEE tests, safety factor of 4 was considered, boards irradiated up to 410 Gy at PSI with no annealing
Further Tests	Some components and the whole bricks were tested in the mixed field of CHARM facility at CERN

JSI: Jožef Stefan Institute (Ljubljana) **TRIGA:** Training, Research, Isotopes, General Atomics
PIF: Proton Irradiation Facility **PSI:** Paul Scherrer Institute (Villigen) **CC60:** CERN Cobalt-60
CHARM: CERN Highly-Accelerated Mixed Field